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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Logistic Regression")
print("-----")
print("Training Samples :", len(X_train))
print("Testing Samples  :", len(X_test))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nAccuracy :", round(accuracy_score(y_test, y_pred) * 100, 2), "%")
sample = [[5.1, 3.5, 1.4, 0.2]]
sample = scaler.transform(sample)
prediction = model.predict(sample)
print("\nNew Sample Prediction:")
print("Predicted Class :", iris.target_names[prediction[0]])

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Logistic Regression
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Training Samples : 105
Testing Samples  : 45

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Accuracy : 100.0 %

New Sample Prediction:
Predicted Class : setosa

```
